

ActInf GuestStream 121.1 ~ Conscious active inference (Wiest and Puniani)

GuestStream | Conscious active inference (Wiest, Puniani) | 1:43:30

Hello, welcome everyone. This is Active Inference Guest Stream number 121.1 on September 23rd, 2025. We're with Mike Wiest, who will be presenting and then discussing OR on Neuronal MTs is the quantum substrate of conscious active inference. So Mike, thank you for joining. Looking forward to your presentation and to the questions people write in the live chat. So take it away, please. Thank you. Thank you, Daniel. Okay. So yeah, here's my title. What I'm going to try and convince you of is, so ORC-OR here stands for Orchestrated Objective Reduction. I'll explain what I'm talking about there and MTs are the microtubules. And so here's an outline.

Uh-oh. There we go. Here's an outline of what I want to talk about. The second part here is the active inference proposal, but I want to start by talking about some evidence to make it plausible possible to you that microtubules may be the quantum substrate of consciousness. Before I go on to the specific active inference proposal, I'm going to mention, I don't think we'll have time to talk about this, but there's another paper I published recently, which addresses some of the more famous conceptual problems with regard to consciousness. I'm not sure if we'll get to those, but I'm happy to talk about those if we have time. But I figured for this venue, we want to focus on the active inference side of the story. And so before I start getting, diving into these details of experimental evidence I want to talk about, let me just summarize the overall proposal we're making with regard to active inference that I'm going to be building up to so that you don't lose the thread of what I'm trying to say. So this is our graphical abstract where I've kind of got two columns here and three rows. At the top is the most abstract level, the theory of active inference that describes optimal behavior of an agent in this mathematical formalism. There is a coarse functional mapping of that theory onto parts of the brain, including details like specific neuromodulators having specific functional roles in the active inference theory. So that match between neurophysiology and neuroanatomy and the theory supports that real brains do use active inference. But then the bottom part is about the implementation. So, so far, most active or all active inference process models, which attempt to say how neurons might implement active inference, this X at the bottom of the picture there by the classical neural implementation is meant to indicate that it seems or at least it has not been established that Hodgkin and Huxley based classical models, meaning classical models that incorporate the actual conductances that generate action potentials in neurons, have not established that those kinds of neurons, those kinds of models, can perform the necessary sophisticated mathematical operations fast enough to implement active inference in real time for, for like a sub-second decision-making and inference and motor control. The other side is our proposal saying that, you know, I want to emphasize that I'm accepting active inference as, at least the first

approximation, a good description of what the brain does. What we're questioning is the implementation of it in terms of classical

neurons and proposing instead that we have to, we have to resort to a quantum model of how neurons work and in particular

taking into account quantum effects in microtubules inside neurons.

So that's the overall proposal that essentially microtubules are going to implement conscious active inference.

But before I come to the active inference part of the story, I want to, I want to convince you that there is at least evidence there's, uh, that microtubules may be

the substrate of quantum consciousness. And I'll start with a, um, for the point of view of how anesthetics render us reversibly unconscious and a bit of experimental evidence from my lab.

But so to, to introduce this part about anesthesia, here's a quote from, uh, Claude Bernard, one of the original investigators when, when we discovered anesthetics and he made this bold connect conjecture.

He said, what is alive must sense and can be anesthetized. The rest is dead. And so for him, the ability to be anesthetized was almost a definition of life itself. And he was basing that on the fact that the same anesthetic gases, we're talking, I'm focusing on volatile anesthetics. Now the same anesthetic gases that render us and other mammals reversibly unconscious also reversibly slow movement or stop movement in invertebrates, single celled organisms with no neurons and even plants. So this chart I've got up here is a, uh, a table from a recent, a review of mechanisms, potential molecular targets of these volatile anesthetics. And you can see people have many candidates, uh, targets, um, in neuroscience,

people tend to focus on ion channels and neurotransmitter receptors, because those are what govern the electrical activity of neurons. But, um, there are other potential targets. And I'm going to focus on the cytoskeletal elements known as microtubules, which we have in common with everything else that is alive. Um, as well as ion channels are present in, in single cell organisms. Okay. So moving forward, um, I want to present this context that I'm comparing two different types of model of what consciousness might be and how anesthetics might reversibly render us unconscious. The classical network, uh, picture is that, um, it's a pattern of neural, uh, of, uh, neural activity, uh, across our brain distributed across our brain. And, uh, something about when it is functionally integrated is what makes us conscious.

Both of these types of models emphasize functional integration over raw neural activity as the neural substrate of consciousness. Um, but the quantum picture suggests that the substrate is act of consciousness is actually the physical substrate of consciousness is actually a physical state of oscillatory state of electric dipoles in microtubules inside neurons. I'm going to tell you a little more about microtubules if they're not so familiar. Um, but these are the two types of models that we're comparing as we looked at the evidence about, um, how anesthetics work. So, um, because these volatile anesthetics bind to many different protein type proteins in the body, contemporary, contemporary researchers tend to

assume that they work by acting on some combination of multiple targets. However, there are some classical

results that suggest instead that they might act primarily on one main target. And, uh, this, uh, first panel here at panel a is showing the so-called Meyer Overton correlation, um, between solubility and olive oil and the, how much of that molecule you need to render someone or an animal unconscious. The solubility in olive oil, um, we, we understand in terms of a, um, hydrophobic binding, a, uh, lipophilic kind of binding. So people think that these, or we know that these molecules bind into, uh, lipophilic or hydrophobic pockets inside many different proteins. Um, but this, the reason that this correlation suggests a one primary or so-called unitary molecular target is that all these molecules that you see in the chart are diverse chemically. Some of them are even just single atoms, uh, elements, xenon and nitrogen are anesthetics at, at, uh, at sufficient concentration. And so the fact that these are chemically diverse, but having, uh, similar functional effects suggests that they are operating not by a chemical lock and key key and lock kind of specific binding, but some more non-specific physical interaction like Vander Waals interactions among protein dipoles.

So that's thing one on this slide panel two or panel B here is showing another fact that suggests, um, it may not be many targets that are relevant, but one main one. And what panel B is showing is the additivity, approximate additivity of doses of different, um, anesthetics. So that half of a dose of benzene plus half of a dose of cyclopropine adds up to about one full dose. And these pairs on this chart are chosen for having different effects on particular ion channels. Like I think isoflurane here has, uh, interact strongly with GABA receptors, the inhibitory receptors in the brain, uh, but cyclopropane does not. But nevertheless, when you add half of a dose of cyclopropane to a half a dose of isoflurane, you get a full dose. So these kinds of data argue against the combination idea, uh, of how anesthetics work. And in fact, well, yeah, I'll leave it there for now. Um, so in terms of what could this one primary molecular target be, you know, I'm going to say microtubules and we have a strong theory, theoretical basis for saying that from this, um, quantum chemical simulation study, which essentially reproduced a, uh, correlation like this, uh, about the binding, uh, in terms of how well the different, about seven of these, uh, anesthetics bind to particular spots on microtubules and disrupt oscillations in those microtubules. Tubulin here are the subunits that make up microtubules. And so this theoretical study essentially reproduces a Meyer-Overton correlation, predicts which molecules are going to be anesthetics and which ones are not by how well they disrupt these oscillations in tubulin. So, uh, that cannot be said for any other candidate molecular.

So, uh, that's a particular target of these volatile anesthetics. So maybe microtubules really are the relevant to functional target. So quickly, what are they? Um, they're, they are literally tubes, uh, inside all cells, uh, not just neurons, but in neurons, they, um, well, and so in a biology class, you first learn that they act, they're called part of the cytoskeleton because they're stiff and they act somewhat like bones. Um, but it is a stereotype to think of them as purely structural because they have many functional roles in single cells and in neurons and in neurons, they have special, uh, organizations and accessory proteins that are different in the axon and the dendrites. And I have this one slide just to stand for the general message that they are not just tent poles in there or railroad tracks. They are participating in all kinds of dynamic information processing and

coordination of movement. So in this beautiful dance of the chromosomes during mitosis, it's microtubules that are choreographing that dance, uh, and actually pulling things apart, doing the movements and in single cells, they are responsible for swimming or, uh, so this is, there's a lot of literature about these kinds of phenomena that I'm not going to go into, but I just, um, this is to plant the idea in your head that we already know that microtubules are not just tent poles or railroad tracks in. Okay. So now coming to, uh, my modest experimental contribution to this story, we wanted to test whether microtubules were our functional target of, uh, anesthetics in this case, um, isoflurane anesthesia that's used in people as well as in animals. And we found, so, um, this drug epothelone B is a drug that binds to microtubules and that gets into the brain, a brain penetrant microtubule binding drug, uh, unlike many of the other microtubule binding drugs that are out there. And when we gave that to our rats, the simple result was they were somewhat resistant to, uh, isoflurane anesthesia as measured by how long it took them to fall unconscious. And so we used, uh, latency to loss of righting reflex as our, as our proxy for time to fall unconscious that the, the righting reflex is if you tip them on their back, if they're awake, they will flip themselves back over. And so when they don't flip themselves back over, that's how you know when they're unconscious. Um, and so here's, uh, our results showing, uh, a, uh, actually a large effect size, uh, uh, 69 seconds longer to fall asleep with that microtubule binding drug than without it. So, um, this by itself does not at all mean that microtubules are doing anything quantum. So we can consider how might microtubules contribute to anesthesia under these two different kinds of pictures. And so under the classical picture, um, one of the important functions of microtubules in neurons is as train tracks for the cute little munchkins that carry synaptic vesicles and other packages to where they need to go inside the neuron. So perhaps, um, anesthetics disrupt that, uh, fast axonal transport, which could plausibly lead to synaptic malfunction, reduce activity levels and reduce, uh, functional integration, which under the classical picture is necessary for consciousness. Um, however, that doesn't seem to be supported by what we know so far. So there are a couple studies showing that isoflurane does affect synaptic transmission, but those studies looked at the mechanism, molecular mechanisms by which it did, and they did not involve microtubules. Uh, in one case, it involved, uh, mitochondria instead. So at present, that picture doesn't seem supported and any, in any case, one might doubt if it would be a fast enough mechanism to account for falling unconscious in a few minutes. Um, before I go on to, to the quantum point of view, let me mention this, uh, cool result in tadpoles showing that, um, uh, they engineered a drug that's not one of these volatile, um, anesthetics to mess with microtubules and, and they were able to reversibly stop their tadpole from swimming around. So this is evidence consistent with the idea that, um, anesthetics might be acting on microtubules and microtubules might be the substrate of consciousness. So here is the abstract from, uh, my RAT paper that I was just talking about. I'm not going to read it, but just focus on this last sentence. This finding, uh, is predicted by models that posit consciousness as a property of a quantum physical state of neural microtubules. So I want to take a moment to say what the motivation for

this quantum point of view is, um, because it's often ridiculed as, uh, explaining one mysterious thing with another mysterious thing. No, the motivation for the quantum view of consciousness is that a quantum model is going to solve a fundamental problem that cannot be solved by a, by a classical model.

And it's a problem that you don't need jargon or math to understand. It's known as the phenomenal binding problem.

Here's a way it's expressed in a neuroscience textbook. Exactly how the parallel streams of sensory data are melded

into or fused into perception images and ideas remains the holy grail of neuroscience.

And I like this dolly painting as a way of, of illustrating that idea. Uh, it's a portrait of his wife where, you know, she's made out of molecules, but somehow, uh, we have these unified conscious experiences.

How do we explain that? And here's how, uh, the famous American psychologist, William James expressed this problem already in his, uh, book around, uh, 1900 or eight, uh, around 1900. He said,

take a sentence of a dozen words and take 12 men and tell to each one word, then stand the men in a row or jam them in a bunch and let each think of his word as intently as he will. Nowhere will there be a consciousness of the whole sentences. So if each one is thinking of a word, there's no sentence there.

Um, his point was this applies to neurons just as much as people standing next to each other.

And the neurons are not even touching each other. There's a synaptic gap between them. So how are, how do we form these unified, holistic conscious experiences? That is what I'm saying a quantum model can do because quantum physics is all about irreducible wholes, W H O L E, like we experience, and there are no irreducible wholes in classical physics.

I'll pause in a moment, but let me

say the proposal. The proposal is that a quantum substrate of consciousness will provide that irreducible unity to account for the unity of consciousness and for its physical efficacy.

So we can talk about that more if you like. But so here's the, the Hameroff and Penrose hypothesis.

I'll go fast about this review of quantum physics. We know

that's a big part of quantum evolution. So there's two parts to a quantum evolution, the local and deterministic part, and then a measurement where you find out what happens. Until then, your wave function only describes probabilities of measurement outcomes.

And people often emphasize the non-deterministic nature of this, but what I want to emphasize is quantum physics is irreducibly holistic, and involves non-local correlations that have been, and what I like to refer to as non-local coordination, which has been experimentally verified at large macroscopic distances of kilometers and resulted in a Nobel prize in 2022. So just illustrating the idea of wave function collapse and pointing out it's essentially a synonym for quantum state reduction. When the distribution of possibilities, we say collapses, and one of those is actualized. So quantum physics has a measurement problem. It doesn't say what a measurement is or what causes the wave function to collapse or when it should collapse. Objective reduction is a theoretical

point of view that says collapse is real, objective, not just epistemic. Okay, so it's a real physical process. And Schrodinger's cat is a fable to say that quantum theory without, if you try to leave out the measurements, that's ridiculous, because then you should be, the theory predicts that there will be half dead cats and those are not observed. So this story that people always talk about is to say, you can't leave out the measurements. So here's orchestrated objective reduction. So objective reduction again, again, refers to a theory in which the collapse of the wave function is a real physical process. And the orchestrated here is referring to in biology, in animals, this objective reduction has been orchestrated so that it corresponds to our, our behavioral problems that we have to solve and helps us to solve them helps to perceive the outside world and so on. So I'm saying in ORCOR associates quantum collapse events in non living matter with moments of so called proto consciousness, it's not until you organize larger chunks of matter like in a brain using microtubules and neurons to orchestrate those that you get something like our, what we call consciousness rather than the proto consciousness.

Okay, so far I haven't showed you much or any evidence for the quantum part of the story.

Here's a hammer off in Penrose, the infamous hammer off in Penrose. Penrose, everyone thinks he's a genius,

but nobody believes anything he says. What's up with that?

Okay, so now I'm going to get to some experimental evidence that microtubules can support quantum effects at room

temperature here in this paper. Let me step back to say the reason this, this, this proposal has been around for 30 years,

the reason people have considered it highly implausible and in fact, we ridiculed it habitually, is that it was when we technologically try to engineer quantum effects. It seems so far that we need things to be super cold. And so people thought the brain is just too hot to support quantum effects, functional quantum effects, functionally relevant quantum effects or macroscopic quantum effects. But now we have direct biophysical evidence in microtubules in vitro showing this phenomenon called quantum phenomenon called super radiance. It's analogous to lasing in a laser. It's at room temperature again. And moreover, the effect gets stronger. That's what this, this plot that's hard to read here is showing the x axis is numbed number of tryptophan residues. And so you can think of that as essentially the size of the structure that you're making out of your microtubule is the basic result on this slide is that the quantum effect gets stronger as you make the microtubules into bigger structures, which is the opposite of the usual prejudice, which says that things as they get bigger, they get less quantum.

So, so far I've, I've shown evidence that microtubules can exhibit a quantum effect, a collective quantum effect at room temperature. Maybe it could happen at body temperature too.

Now, this next slide is just to represent a whole bunch or a whole bunch of experiments and a series of papers from Anirban Bandiopadhiye's lab showing that, so he's measured microtubule, high frequency microtubule

resonances, and he's able to stimulate them as well and show that they predict and control spiking activity.

So this is just a schematic to show he's measuring from multiple neurons in a dish now.

And so without going into details here, the results I want to take are, A, that microtubules interact with membrane voltage and spiking activity. And critically also, their effects span multiple neurons, which is going to be critical if this is going to be the basis of a solution for the binding problem, which is how do we unify neural activities that are happening centimeters apart in the brain. Okay, so I take this to be important experimental work that not many people have heard about.

And finally, for this part about experimental evidence. So here is evidence of a role for microtubules in living neurons. And here is one of a series of papers showing an entangled quantum state in the alive human brain.

This made a little splash when it came out, but I don't hear people arguing about it anymore.

They're just ignoring it. So these researchers just used a standard MRI machine, but dialed it to look for a different kind of signal that they argued would be a signal representing quantum entanglement. They recorded a signal with their MRI machine that looked like an EMG of the heartbeat, which they argued should not be classically observable with an MRI machine. They saw the fidelity of this putative entanglement signal in the human brain correlate with being awake versus asleep when one of their subjects fell asleep in their scanner. And they also correlated it with working memory performance in a different paper. So that's direct biophysical evidence that we have a macroscopic entangled quantum state in our brain, and it's actually related to our conscious function. In terms of people disbelieving this, there is one paper by one author who says he doesn't believe it, but he doesn't provide any alternative mechanism, classical mechanism to explain the signal that they observed.

Here's a good place to pause and see if there are questions or clarifications, objections.

Sure, I'll read one question from the live chat, and then this is awesome. So Alex Sabine wrote, do the 12 people and the sentence integration, gestalt of the sentence, relate to the blind man and the elephant? How?

How?

It's a little different from the blind man and the elephant. So the blind man and the elephant is about how different people are getting different aspects of the reality and none of them see the full picture. This example with the 12 men and the sentence, think of the men as being as representing neurons. And if each neuron represents one feature or word, let's change the example from a sentence to just

seeing an orange chair. The color orange may be represented by neurons in one place in the brain, where the shape of the chair might be represented by neurons in another part of the brain. So we can imagine two men standing next to each other or two neurons, one of them thinking orange and one of them thinking chair shape. That does not add up to somebody having a thought of an orange chair or a perception of an orange chair. That's what that example with the men thinking of a sentence is meant to illustrate, the binding problem.

That also reminds me of the Daniel Dennett thought experiment sequence that starts with the brain and

separates it and continues to separate it and lump and split different components and get exactly at that, which is kind of like, but even this thought example of the 12 people, but already it's presupposed that each of them have a unity. So then how do we approach that challenge of compositionality and unity and multi-composition systems?

Yes. So this issue is, I talk about in more detail in that other paper I mentioned, I'll just flash it up here, this one, where I describe how entanglement, quantum entanglement is the structure of that unity. So the entanglement among different parts of the brain, which I'm hypothesizing, if that can happen, that provides a physical basis for which things are unified and which are not. So in the Dennett experiment, if you start taking out or separating this neuron and that neuron or that brain area and that brain area, if they are no longer able to be quantum entangled, that's when this picture would predict you lose the unity of consciousness. So we don't have to, it doesn't get all fuzzy and philosophical. We can, we have a physical basis for that unity of consciousness under this picture.

Awesome. Carry on.

Okay. So I presented basically three lines of three or four lines with the anesthesia story of evidence, experimental evidence for microtubules and for quantum effects in microtubules related to consciousness. So I'm just going to mention here, there's a large literature that maybe people are not aware of, about so-called quantum cognition, where people's answers to questions or judgments about things can appear irrational from a certain perspective.

So I'm just going to mention here, there are so-called question order effects and logical fallacies that can be explained

if you do a quantum-like model of human cognition. That in itself doesn't mean the substrate has to be quantum.

And this literature is largely agnostic about that or probably assumes it is classic, has a classical basis.

But there's a question, no one has shown you can get those kinds of effects from a classical model yet.

So I'm just pointing to that set of issues and that literature without a lot of discussion right now.

And now to make the transition to the active inference part of the story, I want to mention a paper by Fields et al.,

including Carl Friston and Michael Levin, where they essentially derived that active inference and biology must be quantum physical. I don't see people talking about that paper for some reason.

Maybe I get the impression that the first author, Chris Fields, is the only one who believes it, and the other authors are just down there. I'm not sure. Maybe your viewers know more about that than I do. But so now I'm going to come to the active inference part of the story and suggest that, among other things, this ORCOR, orchestrated objective reduction in microtubules, can account for discrete cycles of perception and can implement active inference.

So here's a little bit of an expanded outline for the active inference part. I guess I will skim the introduction to active inference, but you can tell me if I should slow down for that. But I'm assuming a lot of

this audience has heard about that before. But then I'll spend a little time talking about

ways that it seems implausible that classical neurons, classical neural models, can actually implement active inference fast enough for real-time sub-second inference. And then I'll talk about how ORCOR can solve some of those problems potentially. Okay, so I want to approach this active inference with this quote referring to a quiet paradigm shift. You know, in the last century, we had our Hodgkin and Huxley model, a deterministic Hodgkin-Huxley model of how neurons work. And then we had this quiet paradigm shift from that kind of deterministic signal processing to probabilistic inference and active inference, which is an outgrowth of that. And I'm going to raise some questions about whether that works or not. So, in other words, there's a tension between the deterministic ways that we think neurons work, according to the Hodgkin-Huxley model, and the stochasticity that you need to assume in active inference. And when I say that, I'm aware that the primary kinds of active inference models are deterministic as well, but I'll clarify that in a minute. But so, under active inference or probabilistic coding in general, our beliefs are represented by probability distributions that represent probabilities of different states of affairs being true, probabilities of different facts. And we update those beliefs by sensory observations. And so, here is Bayes' rule way of expressing this updating procedure, where you have a prior belief about states of the world s . And that is compared to some observations of the data that we have to generate an updated or so-called posterior belief based on the mismatch between or so-called prediction error, the mismatch between your prediction and the observation. And these, the diagrams here are illustrating how this could be mapped onto neural anatomy. If I just focus on the panel B here, it's showing three layers of cortex where the primary input from the thalamus that you can think of as sensory input comes into layer four, where those cells get sensory inputs on the purple lines and they get predictions from the black lines. And so, if those red cells can compare the predictions to the observations, that's how roughly we are imagining neurons computing prediction errors, the difference between what you thought was the case and what you observed, that are going to be used to update your beliefs, which again are these probability distributions saying how likely different things are to be true. Okay, so that was a general description of predictive processing as it's called or predictive coding. Inference generalizes that picture to include the agent's behavioral response among the states of the world that the your probability distributions are modeling. And this picture shows a mapping of different variables in the active inference theory onto parts of the brain. I won't dwell on this, but there's an impressive match between variables in the theory and anatomy and also neurophysiology, real neural responses and pathological responses can be accounted for. All sorts of things can be accounted for behaviorally and neurally with this theory. So, it does. So, I want to emphasize that it's not this theory that I'm questioning. It's the classical

implementation.

Okay, given this normative theory or abstract algorithm about how an agent should behave or should infer perceptions or states of the world to be optimal, we can consider how could we get neurons to do that.

And there's basically two broad strategies. I won't go into too much detail, but just make you aware that there's a stream of the literature where neural activities represent expected values like means and variances

of the probability distributions that we're trying to represent.

And the alternative formulation is neural activities are just sampling each time they fire or at each moment, they're sampling

stochastically randomly from that probability distribution that represents your beliefs

rather than representing it in terms of parameters of a parameterizable distribution like a Gaussian.

Okay, and there are pros and cons to these that I'll skip for now.

So, now this inference cycle that I outlined with this Bayes rule here is intractable for most real parameters in our model,

in our beliefs, to optimize our beliefs.

Okay, so that's our first approximation. Before going on to more about the computations that you need need in to implement active inference, let me say how I'm thinking of consciousness.

And I didn't make up this way of defining consciousness, but got it from the literature. But there are a few different ways that people in the active inference literature think of consciousness.

One is in terms of boundary representations.

I won't try to describe what those are because I'm not going to use that definition.

Or this mid-level hierarchical representations might be postulated to be the conscious ones.

Instead, I'm going to use this third concept, which is the most intuitive one, I think, and was endorsed by Carl Friston in at least one paper.

So the idea here is conscious agents, as distinct from

reflex-acting agents or unconscious agents, are ones that have what they call counterfactual or temporal depth.

And that's really just equivalent to a fast implicit planning function. Before each moment of consciousness, the agent needs to integrate over a bunch of possible interpretations and possible plans for what to do next to come up with the optimal inference and perception and next action.

And so that corresponds, this counterfactual word here is referring to considering things that might happen, that haven't happened, that aren't, that are fictitious things that you might do in the future. And temporal depth here is referring to how much into the future you're implicitly considering before making your inference or your decision.

So that's my operational definition of consciousness in the context of active inference.

So I'm going to talk about what a agent with temporal depth needs, what computations it needs to be able to do according to active inference.

First, I want to point out briefly, and then I'll come back to this later, that although we hear it formulated in

terms of the free energy principle, that formulation is equivalent to a formulation in terms of least action, the least action principle.

And under that formulation, you describe the inference cycle in terms of the inference cycle in terms of a path integral, which is just doing what I, basically the implicit planning function that I just described intuitively.

Here, here, X_0 is some state at some initial state, and X_t is your state or position some later time, and the paths are all the different trajectories, actions you could take in between to get to that later state.

And then, active inference is equivalent, or conscious or sentient active inference is equivalent to implicitly considering possible futures to decide on an optimal one.

And again, this is not me making this up.

This is in the textbook, and the literature, it's well established.

But I also show it in the paper that, the pair of papers that we're talking about.

Okay, so now coming to some, some of the things that you need to compute.

Well, maybe I'll pause a moment to see if, if that was too fast about active inference.

Nope. Great, continue.

Okay.

Okay, so we're taking active inference for granted.

See if I can make you see my face.

But I'm questioning a classical implementation.

And so I'm going to focus on a few, a few points listed here.

But first, let me introduce the computations that we need.

So I already mentioned this first approximation in the top line there.

And this green row E refers to another approximation that I'm going to mention later.

I've got a bunch of different computations that are needed to do the path integral.

But I'm just going to focus on the need for linear sums, linear combinations of firing rates of different neurons.

That you need to do that path integral.

And integral is a sum.

So I'm going to focus on that operation and the stochasticity or randomness in the neural dynamics that is also necessary.

And then after that, I'll talk about that problem of discrete perceptual moments that I mentioned earlier.

Okay, so here is an example of an active inference process theory.

Again, what they mean by process theories are hypothetical mappings onto neurons.

And so I'm just highlighting here down at the bottom that what I said before is in the literature, not just me saying it.

It's equivalent to following Hamilton's principle of least action or path integral.

But anyway, the neural models that they use are idealized.

They are firing rate models that don't include spikes.

Some of them do include spikes, but not by modeling the actual conductances that generate the spikes, just as with random number generators.

So firing, another idealization is that firing rates are interpreted as probabilities.

Because remember, we're trying to use these neurons to represent probability distributions.

And in the theory, I didn't show this, but in the equations of the theory, we need to consider logarithms of probabilities as well.

And so to get neurons to do that, they assume that postsynaptic potentials are proportional to logarithms of probabilities

and firing rates are proportional to raw probabilities.

They have an idealized function to translate those potentials into firing rates.

They assume they can add, that different neurons can add their firing rates linearly.

And they assume that conscious inferences happen every cycle of a so-called theta rhythm in the brain.

But they do it by hand in the model without any mechanistic model of how neurons might do that.

So I'm going to consider it now, just because these are idealizations or approximations doesn't mean they're wrong.

But I'm going to point out that they are not established as being right.

And it's somewhat implausible that they are right, that they will, are sufficient to do these computations in fast timescales, sub-second timescales for real time control.

So, a problem I didn't mention on the outline, but it relates to adding together different possibilities of what might happen or what you might be seeing.

This is from a process model describing a simple behavioral task using active inference to describe, to make a neural process model of a simple behavioral task in a rat.

and it had four positions of a behavioral target

and four spatial positions the rat could go in

and that meant they needed a separate neural population

for each combinations of all the different factors

that are involved in the environment

the agent is trying to represent.

In this case, it was 16,

but this points to a potential combinatorial explosion

if you need different populations of neurons

to represent every possible combination of things

that might be out there and things that you might do.

Okay, so coming back to this quote

to highlight the potential tension

between the deterministic signal processing point of view

and the new probabilistic point of view.

The question is,

how do deterministic Hodgkin and Huxley neurons

do probabilistic inference?

So I'm not putting up these equations.

This is the Hodgkin and Huxley equations.

I'm not putting them up to go into details
about what they're saying,
except to show they're all deterministic differential equations.
There's no random numbers over here.
This is a purely deterministic evolution.
Over here, we've got a probabilistic algorithm.
How do you get deterministic neurons to do,
classical neurons to do that?
Well, one of,
so I said one of the operations we have to be able to do
is to add neural activities linearly.
And so here's a paper from 2006
where they modeled neurons
to see if they could get linear summation,
the straight line in this plot.
And in order to get that,
to achieve robust linearity
with these integrate and fire neurons,
they had to average over more than 1,000 neurons
and more than 1,000 trials.
And to get some randomness,
they had to assume randomness in the inputs.
So my point in showing this is that
if you imagine how many neurons would it take
to get robust linear summation like this in one trial,
which we need in order to be able to use active inference
in real time,
it seems heuristically we would need a million neurons.
Too many.
That's on the order of how many should be able to represent
a whole range of different features,
not just one linear combination
pertaining to one of those features.
So this is to say it appears to take too long
and this model is not even a realistic
Hodgkin and Huxley-based model
that incorporates the real basis of the spikes.
But the field of active inference

seems to be relying on this paper
and maybe one or two other models like this
to establish that classical neurons
could do these operations.

So the other issue I said I was going to focus on a bit
is the issue of neural variability.

Real neurons exhibit stochasticity or randomness,
variability in their firing from trial to trial.

Even if you give them identical stimuli,
the variability in stimuli,
the input variability does not account for the neural variability
and neither does the known randomness at synapses.

I'm sorry, I don't have the references for that up here,
but they're in our papers.

So remember I said there were two approaches
to representing probability distributions
in active inference, the deterministic version.

Well, they need to somehow get this randomness
or stochasticity into their model
to match real neurons as we record them
in real behaving animals and people
because it's not there in the deterministic model.

And on the flip side,
the sampling approach assumes stochasticity in their model,
but if Hodgkin and Huxley neurons
are presumed to be underlying that model,
where does the stochasticity come from?

Since that model is deterministic,
and again, the stochasticity at synapses is not enough
to account for the stochasticity of real neural activity.

So these are not to prove
that a classical implementation is impossible,
but to say it has not been demonstrated
with neural models that incorporate
the known Hodgkin and Huxley conductances
that are the basis of the action potentials.

So this is a parallelism.

Also, I mentioned that this formalism for a conscious agent

is equivalent to a path integral,
considering perceptual interpretations
and behavioral responses.
that path integral is also intractable computationally.
And so another approximation is introduced
into the classical process models
called the mean field approximation,
where you assume each of the time steps
of each of these hypothetical trajectories
into the future,
paths or policies of what you're going to do in the future.
You have to assume that each time step into the future
is statistically independent of the other time steps,
which is not really a good assumption intuitively
because our behavioral plans should be coherent plans
that are not random choices at each moment
during that behavior.
So this is not a great thing that we're being forced to do.
But I'm mentioning it because the quantum model
will not need to make this approximation.
So we already saw the need or there was already a need
for multiplication and addition
because the active inference equations
have matrix multiplication in them.
This mean field approximation
implies more need for multiplication
because when you treat different factors as independent,
you multiply their probabilities
to get the overall probability of some path
or trajectory in the future.
And so that leads to a sort of catch-22.
If your neural activities,
if you postulate that your neural activities
correspond to probabilities,
then maybe it's relatively straightforward to add them,
but multiplying them becomes non-trivial.
And the flip side, if you postulate
that neural activities correspond

to logarithms of probabilities,
then you just have to add them
and that is equivalent to multiplication.
But then if you need to add raw probabilities,
that becomes non-trivial.

So it seems like whichever way you choose,
you have to leave out something that you need.

And so the solution that's been adopted
in the example process model there
is that maybe the synaptic potentials
are representing the log probabilities
and the rates are the raw probabilities,
the firing rates of neurons.

We have some experimental evidence.

I won't dwell on this experiment,
just showing that in the abstract,
they're showing that neural activity
in this area LIP of a monkey
was consistent with its integrating
a variable that was proportional
to a log likelihood ratio,
a log likelihood.

In other words, a log probability.

So there's some experimental evidence
that neurons do compute log probabilities,
but that's not equivalent to a classical model
that shows how a classical Hodgkin-Huxley neuron
can do it.

To my knowledge, we do not have such a model.

Similarly, these models tend to almost universally
rely on convergence into attractor states,
which are quasi-stable patterns of activity
that represent different possibilities.

And this has been criticized by people other than us
because when we're talking about fast real-time inference
as opposed to learning on longer timescales,
for fast real-time inference,
attractors might be too slow

because different areas have different conduction delays
from the eyes or the ears to the cortex,
to the visual or auditory cortex.
And sensory inputs are changing from moment to moment.
So it's not so easy, potentially,
to get a quasi-stable pattern of activity going.
And there's a question about how long
do you have to keep it going
in order for it to register as that's the answer.
So that's another doubt about whether
classical models can account for these computations.
So here's a summary of that critique
of the classical implementations.
And in the title there,
I should have been more specific than realistic.
I mean, by realistic, I mean a classical model
that incorporates the Hodgkin and Huxley conductances
that are responsible for action potentials.
So what these models,
these classical process models have not established
is that they have not established
that Hodgkin and Huxley neurons
can quickly add their firing rates linearly
in like 200 milliseconds.
The fluent alternation that I was talking about,
whoops,
between multiplication and summation
that is necessary for matrix multiplication
not been demonstrated.
The log,
some variables are logs of probabilities
and some are probabilities.
How do you do that with neurons?
There's no,
I don't know of a model that shows
how to do that with realistic neurons.
The papers that did show linear summation
with neurons suggested it would take too long

or too many neurons
to get addition to work right,
to do this in under a second.
And here I mentioned there's a problem accounting
for the real stochasticity or variability in neurons
given a determinative model.
So I next want to go to the quantum picture.
I guess I'll flash one more thing
before I pause to ask for questions,
which is as a sort of segue to the quantum proposal.
I mentioned earlier this paper from 2022.
Based on these results,
at the bottom of the abstract here,
we suggest that biological systems
employ quantum coherence
as a computational result,
as a resource,
and implicitly as a communication resource.
And here is a place
where they're very,
sort of cryptically saying
they think perception is quantum.
One might imagine
that a combination of quantum computing
and neuromorphic engineering
may be able to account
for real human vision.
The results obtained here
suggest that quantum biology
is far larger than we thought so far,
and all biological systems
are quantum systems
and use quantum coherence
as an information processing resources.
This has been suggested before,
but here we showed
that it follows from general principles.
That's what I underlined

at the bottom there.

Sounds like a big deal,

but I haven't heard anybody else
talking about it.

Okay, so before I go on,

any questions,

clarifications,

complaints?

There are a few questions.

I think continue on and close out,

and then we'll all read them

in the discussion.

Thank you, though.

Very good.

Okay.

Okay, so I talked about

some problems

with particular computations,

and now let me introduce

another issue

that any theory

of conscious active inference,

any process model,

has to address,

which is that

our moments of experience

are non-overlapping discrete moments,

like frames of a movie.

And I don't,

this does not seem to be

largely appreciated,

but it is well established.

So the kinds of experimental evidence

that establish this

are, for example,

the large psychophysical literature

on backward masking.

So the bottom schematic there

is illustrating a trial like that,
where the schematic is showing
what subjects would be looking at,
they're fixating
on the dot in the middle.

The pair of bars here
of unequal length
are flashed very briefly,
and they're supposed to note
which one was longer,
and that's to test
whether they saw those bars or not.

But after those target bars
are flashed very briefly,
these sideways equal signs
are the masks
that are flashed briefly
after the targets.

and if you flash them
soon enough,
within a few hundred milliseconds
of the targets,
you can prevent the targets
from being seen at all.

And so a large literature
of effects like that
appear to prove
that our brain integrates
some finite duration
of time
to generate
each of our moments
of perception.
perception.

And, you know,
you can reflect that
if that weren't true,
we wouldn't be able

to perceive movement
because movement
we perceive
by noting something
was in a different place
at two different times.
So there are many,
there are a bunch
of other experiments
that come to a similar conclusion.
I think I'll,
I won't describe them now
for time,
but I think
the backward masking
is,
is enough to establish
that
these cycles
of perception
are non-overlapping.
So it's not like
a continuous update
where
new sensory information
kind of breaks thin
whenever it gets,
whenever it's
observed.
Rather,
there are,
you only get
to be noticed
at these
discrete moments.
And this is
also a requirement
in conscious

active inference.

I think there's

a little bit of

confusion about this

maybe in the literature,

but it's clear

that even when you go

to a continuous

description

of active inference

using differential equations,

you still need

to supplement that

with,

if it's going to be

a conscious,

temporally deep

agent,

you need to

supplement that

continuous,

instantaneous description

with something

that integrates

over a finite

window,

the temporal depth

to generate

the sentient

or conscious

inference.

So,

we know

in real life

there are

discrete moments

of experience

and conscious

active inference
requires discrete
moments
of experience.
The question
is,
how do you
get neurons
to generate
discrete moments
of experience?

Before I go
to that question,
here's just
another way
of seeing
this result.

We can see
neural correlates
of these
moments
of the movie
of our life
during binocular
rivalry experiments
where you show
two different
pictures
to the two eyes
and instead
of seeing
a combination,
people alternate
seeing one
or the other
and the stimulus
is not changing.
Both eyes

are still
seeing the same
picture the whole
time,
are being shown
the same picture
the whole time,
but your brain
decides what
it's going to see
at each moment
and it alternates
between the two
pictures.
and what
this study
is 2009
study showed
with EG
electrical recordings
from the scalp
is moments
of neural
synchronization,
rhythmic moments
of neural
synchronization
that correspond
to these moments
of perceptual
experience.
Okay.
Okay.
So,
recall this
was the
ORC-OR
hypothesis

and
here again
is the statement
of the idea
quantum collapse
events
is what
OR refers
to.
It doesn't
become consciousness
until the
objective reduction
events are
orchestrated
by biological
structures
to be more
complicated
and informative.
Now,
coming back
to the
path
to integral
formulation,
I want
to emphasize
the distinction
between the
path integral
formulation
in quantum
physics
and in
classical
physics.
You can

formulate
classical
physics
and active
inference
in terms
of the
least action
principle
which is
expressed
in terms
of this
path integral
but in
classical
physics
that's
optional.
it's
redundant
with the
instantaneous
description
in terms
differential
equation
kind of
description.
It's
equivalent.
The system
just follows
the path
of least
resistance
they say.
The situation

is different
in
quantum
physics
that's why
I say
here it's
mandatory.
You can't
get away
with an
instantaneous
differential
description.
instead
and I
want to
also say
this path
integral
formulation
of quantum
physics
is the
most
fundamental
formulation.
Things like
the non-relativistic
Schrodinger
equation that
are introduced
as postulates
in introductory
courses can
be derived
from the
path integral

formulation
in the
non-relativistic
limit.

So the
general
description
of quantum
physics is
in terms
of this
path integral
where at
each moment
nature
decides
what it's
going to
do next,
what it's
going to
do at
the next
quantum
measurement
by considering
every
conceivable
trajectory the
system could
take in
between now
and that
future
measurement.

It
integrates
over all

those
possibilities
without
any
approximation
like we
had in
the
classical
case,
the mean
field
approximation.
And in
the quantum
case,
again,
this is not
equivalent to
the
instantaneous
description in
terms of
forces or
derivatives.
that's why
I'm saying
it's a
mandatory
description.
And so
it's as if
nature
considers
everything that
it could do
next at the
next measurement

event and
picks an
optimal one
based on
minimizing the
action,
the physical
action.

That's the
least action
principle that
I said is
equivalent to
the free
energy
principle.

And that's
why our
proposal that
orchestrated
quantum
objective
reduction
could
implement
conscious
active
inference
is
founded on
a rigorous
equivalence,
mathematical
equivalence,
not a
loose
analogy as
one of our

reviewers
called it.
Maybe I'll
pause there
because this
is the
proposal.
Before I
flesh it
out,
this is the
proposal.
if you
want to
clarify
anything.
Indeed,
that there's
a path
that can be
splined or
interpolated
that's based
upon the
biological
time cycle
and
cognitive or
computational
capacity and
frequency of
actual
mechanistic
processes that
are known
where
necessity
and

sufficiency
and
non-human
models and
relevant human
data exist
to come
together to
provide a
mechanistic
how along
with the
kind of
full
MAR
levels of
analysis or
even more
of like a
Tinbergen
style
four questions
analysis.
That's what
theory
integration looks
like in
biology.
That's where
it's at.
Okay,
so yes,
on this
slide,
I'm at
the
algorithmic
level and

I need to
specify what
is going to
implement it
at the
bottom level,
the implementation
level.

And I'm
saying
microtubules
can do
that.

but this
formalism
applies to
nature at
large in
general.

Nature does
this and
we're saying
it does it
in a
particular way
in brains
where it's
been orchestrated
so that
what is
represented is
something that's
useful to the
organism.

Here's just to
make it a little
more concrete
how I'm

picturing this
happening in
neurons.

So these are
cartoon neurons
where I've got
these fat
dendrites with
microtubules in
them and
here are some
axons with
synaptic terminals
and I'm
pointing out
there's a lot
of actin in
the synaptic
terminals and
actin can
exist in a
liquid crystal
state which
is used as
logic gates
in optical
computers and
we've got a
quantum optical
I'm hypothesizing
a quantum
optical computer
in the
microtubules
so the
actin at
the
synapses

could be
acting as
gates that
modulate
synaptic release
probabilities
based on
the quantum
state in
the
microtubules
and the
quantum state
in the
microtubules
reflects neural
activity on
the membrane
because there
are voltage
gated calcium
channels all
over the
neural
dendrites and
microtubules and
microtubules
associated proteins
are sensitive to
calcium and so
that's how so a
critical requirement
for this hypothesis
to be viable is
that this
hypothetical quantum
state in the
microtubules must be

able to communicate
with action
potentials on the
membrane both ways
because we know
action potentials is
how our sensory
inputs are
represented and
we know action
potentials is how
we act
contract muscles to
behave including
talking so the if
this is true the
microtubules have to
be able to talk to
the membrane
potential and the
way we can imagine
that concretely is
that the way the
membrane potential
talks to the
microtubules is
via calcium the
pattern of
intracellular calcium
mirrors the pattern
of electrical
activity across the
membrane so it can
imprint that onto
the microtubules
oscillatory state
that's how the
microtubules find out

what you're looking
at and now how do
the microtubules
modulate what you
do in response by
modulating synaptic
release probabilities
at the axon
terminals where we
know where in the
central nervous system
synaptic vesicle
release is notoriously
unreliable they say
noisy and they
speculate about why
would you want
unreliable neurons
well that provides
a point where the
microtubule state can
modulate the neural
activities without
violating the known
Hodgkin and Huxley
dynamics that we know
about
so it's interesting
that that's also
called quantized
release and it
really does come
down to the to
the discrete
packets of vesicles
themselves with
discrete composition
of of molecules

and neurotransmitters
that's true that
could muddy the
issue I'm not
claiming that
discrete packets we
do call them
quanta but they
are not quanta in
the quantum
physical sense
necessarily
yeah but yeah
it is it is kind
of funny that we
use that word
I think yeah I
want to emphasize
that this gives
an advantage to
beings that can
involve such a
conscious dynamic
whereas consciousness
appears epiphenomenal
and useless in
classical models
the capacity so
there's a tendency
to say quantum
computation is hyped
overhyped because
you know there's
only a few
quantum algorithms
and they're mostly
useful if you're
trying to rob a

bank
so maybe it's
all hype but
speed is not the
only potential
advantage of
quantum computation
memory capacity
is another one
and this is a
significant issue
for neuroscience
because our
models of
neuro of memory
are linear
models like the
Hopfield model if
you've heard of
that are their
capacity how many
memories you can
store is linear in
the number of
neurons so if 10
neurons can store
two memories 100
neurons might only
store 20 memories
and it goes up
linearly
that's a severe
constraint and
there's a question
about whether that's
enough to account
for our massive
memory capacity

over decades and
100 years
in contrast
by marrying
the quantum
algorithm search
algorithm called
Grover's algorithm
with a neural
network architecture
these guys in
2000 came up
with a quantum
associative memory
whose capacity
was exponential
in the number of
neural units
since then
there's been
developed further
development of
related models
where the
advantage is
polynomial or the
capacity is
polynomial or even
exponentially better
than the classical
models so
there's potentially
a huge
memory advantage
if we're just
looking for the
bottom level
what would

what would it
how would it help
to have such a
quantum system to
use in your brain
potentially it can
account for our
ability to remember
things over a
hundred years and
use single
episodes from
decades ago in
our momentary
inferences about
what to do in
sub-second time
scales

I don't have a
slide for it but I
want to briefly
mention that there
is also a
concrete model
by Behera and
Elitzer in
2005 showing
how you can
program neurons
it's basically a
quantum process
model how you
can program
neurons assuming
a quantum state
inside the neurons
which they don't
speculate about what

it might do but
when they assume
that they get
eye tracking
behavior that
dramatically
outperforms the
classical common
filter type models
and introduces
discrete jumps in
the eye movements
when you're trying
to follow a moving
target with your
eyes that match
real qualitatively
match real eye
tracking data but
are not present in
the classical models
so that was back
in 2005 I don't
specialize in that
field but I'm not
aware of any newer
developments or
models superseding
those so because
the quantum
description is
equivalent to the
pathog integral
formulation as I
tried to emphasize
earlier that
quantum model is an
example of how you

could neurally use a quantum system for active inference for a concrete behavior of tracking something with your eyes and so that simple model could potentially be generalized to more complex behaviors.

Okay so there's that model illustrating concretely how you could use this in a brain I'm saying there's also the potential memory advantage of a quantum model and so given the experimental evidence that I reviewed at the beginning that's why I say quantum collapse on the microtubule state inside of neurons is actually experimentally supported it's consistent with what we know about neurophysiology in a way that arguably these classical models are not even though I admit this is a speculative model it is consistent with the deterministic function Hodgkin-Huxley function of the neurons and doing the

quantum modulation through
the stochastic synaptic
release as a potential
example.

So here I'll wrap up now
this table just summarizes
different issues for
theories of consciousness
the classical theory or
quantum theories I've been
emphasizing issues around
probabilistic optimization
and active inference and
this discrete decisions
moments issue but I
brought up the unity of
consciousness the binding
problem epiphenomenalism I
didn't even mention affect
whoops or non an arrow of
time subjective arrow of
time but these are some
other issues where the
quantum model may have an
advantage so I'm just
throwing those up there for
completeness.

So I think to finish I'll
just put my outline so you
can point at stuff that you
would like to ask about or
talk about more.

Awesome.

Okay there's a bunch of
written questions and anyone
else can write some
questions while I just
compile them and move into

the discussion just maybe a little bit how did you come to work on the empirical work and the animal biology and the theoretical biology and just how does that come to be that way in 2025?

Yeah well I got interested in this idea way back in the 90s when I was a physics grad student high energy theory elementary particle physicist so I got excited about these ideas and got my foot in the door to neuroscience through computational neuroscience and then decided I needed to learn to do experiments so I wouldn't be begging experimentalists for their data to analyze and I learned to do multi-electrode recordings in behaving rats and my motivation there was interest in the binding problem to address that from the main from a classical neuroscientific point of view you want to be able to look at multiple neurons at the same time and see how they're coordinating in real time or synchronizing but it's not till the past few years that I've managed to join the discussion about the quantum hypothesis by I noticed that so these microtubule binding drugs have been known for a long time as anti-cancer drugs

because they interfere with mitosis
and cancer is runaway cell
division and that is governed by the
microtubules so messing with the
microtubules apparently helps to
slow down cancer but most of those
drugs as I mentioned don't penetrate
well into the brain but more
recently I came across a review
describing people developing brain
penetrant microtubule stabilizing
drugs to go after brain cancers and
that made me think maybe that's an
opportunity to test whether
anesthetics target microtubules
awesome all right I'm gonna ask a
bunch of questions that have been
written in the chat there they're all
over but it should be fun and and
associative all right first your
colleague Ernesto Soros wrote oh is
there an analogous phenomena for
qualia to the collapse of the wave
function in other words is the
collapse of some wave function in the
brain what makes us aware of a qualia
yes in in under the oracle our theory
we're equating moments of consciousness
with moments of orchestrated collapse and
so the way I would phrase it to try and be
philosophically careful is the
consciousness the conscious experience is
a property of that physical collapse
process and the collapse is when the
irreducible unity or coordination or
non-local
correlations manifest themselves and so
that's how it can be the physical

objective basis to the unity of
consciousness that we experience
great transition to the next question
from what you hear from Ernesto hi Ernesto
there's more coming Benjamin G wrote
if the Penrose Hammeroff work OR model
is a helpful model I wonder if that
increases the likelihood of the fuller
Bohm Penrose Hammeroff model and other
David Bohm isms not just Penrose isms
I have not heard of a Bohm Penrose model
because I think the Bohm model is not
consistent it is a different model that's
not consistent with orchestrated with
objective reduction now there are a
number of different theories of
objective reduction and Penrose has a
particular one but the Bohmian
interpretation or theory of quantum
physics and my understanding does not
have actual collapse you just find out
what happens it still has the non-locality
though so in that respect it might still
provide a basis for binding but in this
picture this picture is totally distinct
from the Bohmian picture
interesting yes with the implicit order and
the pilot wave

okay

Benjamin asks

my understanding is that tubulin and
microtubules are not only in human or
mammal neurons but e.g. are similarly found in
fungal hyphae and you know editorial note
all all cellular life so what are implications
of this on fungal computation
fungal um i am not prepared to talk about
fungi but i i have learned about single

celled organisms i mean yeah almost all life has microtubules i think there are some exceptions and even they may have microtubule like proteins but speaking broadly it's yeah cellular life tends to have microtubules um and in line with that i mean that's related to my point earlier that volatile anesthetics that reversibly make us unconscious have analogous effects at comparable concentrations on single cell paramecia that swim around without any neurons but they do have microtubules and they use them to swim with um so this picture i would say unlike the classical picture that emphasize that is all about the skin only of neurons or the membrane of neurons only this picture makes it understandable how our consciousness is continuous with similar stuff earlier in biology you know um what i don't know if a paramecium is conscious but it appears to be it's it's sensitive i've done it myself i've anesthetized paramecia with isoflurane and i'm trying to do the same experiments that we did in rats on tetrahymena and um paramecia single cells to implicate microtubules in uh the anesthetic effect on them but um what i'm suggesting is that evolutionarily microtubules already had a function of integrating sensory information and coordinating movement we've just built on that maybe also just um there are microtubule stabilizers and destabilizers you know different mechanics of how they bind different places etc so dissolving seems to remove their function but also you talked a lot about microtubule stabilizing which i know over longer time scales can influence the pool of subunits and the assembly dynamics even through the stabilizing kinetics but in the shorter term is that what is being hypothesized to matter or like is it about them being obviously again breaking them down just dissolves the substrate so that's that's easy for a loss of function on on transport as well as any any other features but does stabilizing remove the function or does it modify you know what what matters the phosphorylation state just their their sheer assembly yeah thanks for asking that question um and some new results have made me question how i was thinking about this but the way i was thinking about this

the stabilization i think was kind of a i thought was a red herring that is um our experiment was just showing that the binding by this drug was maybe getting in the way of the isoflurane and so it's it's certainly not that we're completely dissolving or depolymerizing the microtubules maybe well yeah so the drug is the stabilizer um so maybe that could have interfered with transport without actually breaking them apart under a classical picture but under the quantum picture it's interfering with their collective vibrations um

okay but so the way i was interpreting our result was simply that it took the rats longer to fall asleep when we gave them the microtubule drug because the microtubule drug was getting in the way the isoflurane so it was kinetically blocking a a in a sort of non in a way that there's also like gpcr binding drugs or or molecules that will sterically hinder a psychoactive so they'll reduce the effect of an ion channel but not have a direct phenomenological effect or even a kinetic effect and so something similar that that sterically hinders something that would interfere with reducing the the quantum action right that's how i was interpreting it but since then i've seen some results so under that interpretation you kind of would expect that the more the the microtubule binding drug you would get uh that you gave the more resistance to the anesthesia you would get however um this is preliminary now but i i i saw a paper that did this in mice and when they gave the drug for two weeks we just gave it one one time and looked what happened the next couple days when they gave it for two weeks they saw an opposite effect where

the mice were more sensitive to anesthesia and i think i may have seen something like that in my single cells too when i left them in the microtubule drug for multiple days um that they became more sensitive instead of less sensitive so that's making me question the just steric blocking interpretation and maybe it does have to do with how stiff the microtubules are and how well they vibrate or something like that um it's it's stories getting interesting yeah there's there's a lot going on there's phosphorylation and there's uh there's a lot going on down there but i don't think it's i i think it's more of a direct or like the modeling study and other uh considerations lean me away from things like phosphorylation to just direct physical interactions like van der waal's induced dipoles interfering with the the dipole oscillations directly it that's a detail you know it interesting of course we're just raising possibilities but for for years i have seen that ideas that the combinatorics of the phosphorylation states could also it's and and this kind of relates to a uh one one question that was in the chat of uh ernesto what what about electrical stimulation of the cortex which elicits repeatable qualia does it really but do different areas of the brain become transiently entangled and then just one one thought that i wrote in there was seems like there could be multilateral causality electromagnetic quantum material changing one can change others so the the fact that there's experiments where one is upstream causally of another when ultimately it's part of the same causal nexus like if you were to deplete nitrogen that would influence microtubule dynamics over some time frame but that doesn't mean that nitrogen plays a proximate mechanistic role but so everything can kind of re-enter the same causal nexus and that's why it is important to do the experimental work that you're discussing okay so

the question remind me the question now again um what about electrical stimulation of the cortex which elicits i think i wanted to okay i think before i comment on that i wanted to you said um um you mentioned the microtubule associated proteins and how they their states are changed by phosphorylation and so on and so i think it might be helpful for me to bring up the metaphor of a guitar where the microtubules are like the guitar strings and you can imagine microtubule associated proteins being placed on like fingering different frets and that's how you get different chords playing which represent different perceptions or experiences and that's how you could store them as memories as well if you can hold on to those frets for a long time and then the actual strumming i guess stretching the metaphor a little bit is the um the uh calcium coming in to affect the the microtubule state and and uh encode the neural activity state into the microtubules um okay so that was just to relate to the microtubule associated proteins and how they could be involved in learning for sure there's there's evidence that they are involved in learning um as far as micro stimulation yeah um under this picture if you activate those neurons and you have a conscious experience because of that it means the the those microtubules got excited

into a coherent state such that they uh a coherent state such that when they collapsed you experienced it um and i was going to relate that to something else but i forgot

oh all good um maybe in sort of close oh i'm sorry i remember so i'm just gonna relate that to how um in the eeg binocular rivalry experiments i i flashed up for a moment there that sort of illustrated the moments of conscious experience as moments of synchronization across the brain a detail i didn't mention is that the synchronization at each moment incorporates neurons that are representing what you are perceiving at that moment so if you're at that moment you perceive a house then your house area becomes active at that moment and enters the under the quantum picture enters the coherent the entangled state or under the classical picture or both pictures it enters the synchronized state that is the neural correlative consciousness as already measured um yeah and when you do it artificially you're just inducing the same state neural correlate neural correlate neural causate who's on first neural substrate i prefer neural substrate to neural correlate

yeah well in sort of closing uh what are your empirical and theoretical directions what milestones or kind of landmarks

or reductions of uncertainty could be ahead how could people learn about this area or get involved or just take it from here yeah so uh definitely needs more testing um and so avenues for testing one avenue that more people seems to me should be trying is imitating the mri experiments of care skins and perez it only it takes a regular scanner you just have to turn the dials in a different way than you normally do and so you could try uh microtubule binding drugs or other kinds of drugs to try and um isolate the molecular basis of that entangled state that they appear to have measured um another experimental avenue would be to try to do spectroscopy experiments on living brain tissue

which people have already done where you'd be looking for peaks that correspond to microtubule resonances and you would be looking to see them either get reduced or shifted or something under anesthesia or other drugs that affect consciousness that although um people have done experiments like that it will be technically challenging because the signal i think will be small in a large background signal but that's a possible avenue um i'm trying to replicate our rat results in uh mice and we think we see that happening in mice and this week i think i saw it happening in tetrahymen a single-celled swimmers which was quite exciting epic any last words um i hope that this uh hypothesis is starts to be discussed more seriously by serious mainstream recent researchers now especially since the uh authors of active inference carl fristen has written a paper saying active inference must be quantum yes we had a 2023 course on physics as information processing with quiz fields and have had a series of discussions on some of those quantum active inference and bayesian mechanics papers yeah there's there's so much there and how exactly that theory becomes uh improved in its accessibility rigor applicability how the broader cultural paradigm etc meta science environment continues to evolve all things of interest absolutely absolutely i'll close with with a blake ism if the doors of perception were cleansed if the doors of perception were cleansed everything would appear as it is discreet excellent in the same poem he said um what is now proved was once only imagined amazing thank you mike good luck and and hope people enjoyed that thank you bye take care thank you

thank you

Thank you.